

Optimizing Domain-Specific Text Generation:

A Comparative Study of Parameter-Efficient Fine-Tuning and Sampling Strategies

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Large language models possess astonishing capabilities across diverse natural language tasks; however, their adaptation to specialized domains remains resource-intensive for most researchers and organizations. Full fine-tuning of models containing hundreds of millions of parameters requires substantial computational resources, creating barriers to customization for domain-specific applications in medical, legal, and scientific contexts (Vaswani et al., 2017; Radford et al., 2019).

Parameter-efficient fine-tuning techniques, particularly Low-Rank Adaptation (LoRA), offer promising solutions by reducing trainable parameters by over 99% while maintaining model performance (Hu et al., 2021). However, the interaction between LoRA hyperparameters and generation quality remains poorly understood, particularly regarding the trade-offs between computational efficiency, text quality, and safety considerations.

The stochastic nature of text generation introduces additional complexity through sampling strategies that fundamentally affect output quality, diversity, and coherence. While techniques such as nucleus sampling have shown promise in addressing repetition and degradation issues (Holtzman et al., 2020), systematic comparison of sampling strategies in the context of parameter-efficient fine-tuning remains limited.

This research addresses three critical gaps: (1) optimal LoRA configurations for balancing parameter efficiency with performance, (2) comparative analysis of sampling strategies for maintaining generation quality, and (3) practical deployment considerations, including toxicity mitigation and model interpretability. The investigation provides empirically validated guidelines for efficient, responsible LLM adaptation applicable to real-world deployment scenarios.

Methodology

Model Architecture and Dataset

GPT-2 Small (124 million parameters) served as the base model, selected for its well-documented architecture, widespread adoption in research, and computational feasibility (Radford et al., 2019). The WikiText-103 dataset provided training data, comprising 10,000 training samples and 1,000 validation samples, offering sufficient linguistic diversity while maintaining manageable computational requirements.

Parameter-Efficient Fine-Tuning with LoRA

Low-Rank Adaptation (LoRA) was implemented to enable efficient fine-tuning by injecting trainable low-rank matrices into attention layers, specifically targeting the attention projection layers (`c_attn`) of the transformer architecture (Hu et al., 2021). This approach reduces the number of trainable parameters from 124 million to approximately 0.12-0.47% of the original, depending on the rank configuration.

Four systematic experiments varied LoRA hyperparameters:

- Experiment 1: $r=8$, learning rate= $3e-5$, $\alpha=16$ (294,912 trainable parameters)
- Experiment 2: $r=4$, learning rate= $3e-5$, $\alpha=8$ (147,456 trainable parameters)
- Experiment 3: $r=16$, learning rate= $3e-5$, $\alpha=32$ (589,824 trainable parameters)
- Experiment 4: $r=8$, learning rate= $5e-5$, $\alpha=16$ (294,912 trainable parameters)

Training was performed using FP16 mixed precision with a batch size of 8 (achieving an effective batch size of 16 through gradient accumulation), three epochs, 100 warmup steps, and a weight decay of 0.01. Experiments were done in Google Colab Pro with Tesla T4 GPU (15GB), requiring approximately 8 minutes of training time per configuration. The

HuggingFace Transformers library and PEFT (Parameter-Efficient Fine-Tuning) framework provided implementation infrastructure.

Sampling Strategy Evaluation

Using the optimal LoRA configuration, eight sampling strategies were evaluated:

- Greedy decoding (baseline)
- Temperature scaling (0.7, 0.8, 0.9)
- Nucleus sampling / top-p (0.9, 0.95)
- Combined approaches (temperature + nucleus)

Diversity metrics included distinct-1 and distinct-2 ratios (measuring lexical variety) and maximum repetition counts. Generation length was consistent at 100 tokens across all experiments to ensure fair comparison. Statistical significance was assessed using paired t-tests with 360 test prompts, providing vigorous validation of observed differences.

Safety and Interpretability

Toxicity detection utilized the Detoxify library, enabling the real-time assessment of generated content safety. Attention visualization utilized BertViz to analyze token-level attention weights across all 12 transformer layers and 144 attention heads, providing insights into the model's decision-making processes and information flow patterns. These explainability tools enhanced transparency and supported error analysis throughout the development.

Evaluation Metrics

Primary evaluation metrics included:

- Perplexity (model uncertainty measure)

- Distinct-1 and Distinct-2 (lexical diversity)
- Maximum repetition count (generation quality)
- Toxicity scores (safety assessment)
- Generation time (computational efficiency)

Results

LoRA Hyperparameter Optimization

Systematic experimentation revealed that rank $r=8$ with a learning rate of $3e-5$ is the optimal LoRA configuration, achieving a perplexity of 31.89 with only 294,912 trainable parameters (0.24% of the base model). Table 1 presents the results across all LoRA experiments.

Experiment	Configuration	Trainable Params	Perplexity
1 (optimal)	$r=8, lr=3e-5$	294,912	31.89
2	$r=4, lr=3e-5$	147,456	35.53
3	$r=16, lr=3e-5$	589,824	33.26
4	$r=8, lr=5e-5$	294,912	32.59

Table 1: LoRA Hyperparameter Experiment Results

Critical findings emerged from these experiments. First, rank $r=8$ achieved optimal balance between model capacity and generalization, with lower ranks ($r=4$) suffering from insufficient expressiveness (perplexity: 35.53) and higher ranks ($r=16$) exhibiting overfitting despite doubled parameter count (perplexity: 33.26). Second, learning rate displayed significant sensitivity, with $3e-5$ producing superior convergence compared to $5e-5$ (perplexity: 32.59). These results confirm that parameter efficiency and performance are not

mutually exclusive, with appropriate hyperparameter selection enabling a dramatic reduction in parameters while maintaining model quality.

Sampling Strategy Comparison

An evaluation of sampling strategies revealed that nucleus sampling with top-p=0.9 is optimal, achieving superior diversity while preventing repetitive generation patterns. Table 2 presents quantitative results across all evaluated strategies.

Strategy	Distinct-1	Distinct-2	Max Rep
Greedy (baseline)	0.347	0.375	5
Temperature = 0.7	0.854	1.000	2
Nucleus (top-p=0.9)	0.949	1.000	2
Nucleus (top-p=0.95)	0.870	1.000	2
Combined (temp=0.8, p=0.95)	0.745	1.000	3

Table 2: Sampling Strategy Comparison Results

Results showed severe limitations of greedy decoding, which exhibited severe repetition issues (maximum repetition count: 5) and extremely low lexical diversity (distinct-1: 0.347, distinct-2: 0.375). Nucleus sampling with top-p=0.9 achieved best performance across all metrics, producing diverse, coherent text with minimal repetition. Statistical analysis using paired t-tests confirmed significant differences ($p < 0.01$) between nucleus sampling and baseline greedy decoding across all diversity metrics.

Toxicity Mitigation

Integration of toxicity detection enabled safety evaluation across all experiments. Analysis of 360 generated outputs revealed that the fine-tuned model with nucleus sampling achieved a 71% toxicity reduction compared to baseline greedy decoding, proving that

appropriate sampling strategies contribute significantly to safe text generation. Average toxicity scores remained below 0.001 for nucleus sampling configurations, substantially lower than the 0.003 observed with greedy decoding (Bender et al., 2021).

Attention Visualization and Interpretability

Attention analysis using BertViz revealed hierarchical processing patterns across the 12-layer transformer architecture. Early layers exhibited strong diagonal patterns, reflecting causal masking and positional encoding. Middle layers demonstrated vertical patterns, indicating a focus on key contextual tokens. Late layers showed complex semantic relationships, including syntactic dependencies and topic tracking. This visualization confirmed specialized attention head behavior, with some heads attending primarily to positional cues. In contrast, others captured longer-range semantic structures, enhancing model transparency for stakeholder review and debugging processes.

System Integration and Deployment

The final implementation integrated all optimizations into a comprehensive Gradio dashboard featuring four interactive components: (1) text generation interface with side-by-side model comparison and adjustable sampling parameters, (2) performance analytics visualizing experimental results, (3) attention-based explainability tools, and (4) a comprehensive experiment results matrix. The system achieved consistent generation times of approximately 1.28 seconds per 100-token output, demonstrating practical deployment feasibility. The interactive dashboard successfully showcased research findings while providing a functional demonstration of parameter-efficient fine-tuning capabilities.

Discussion

This research demonstrates that parameter-efficient fine-tuning using LoRA provides a viable path for broadening access to LLM adaptation, reducing computational barriers while maintaining model quality. The identification of rank $r = 8$ as the best aligns with theoretical predictions from linear algebra regarding the effective subspace dimensionality for low-rank approximations (Hu et al., 2021). The observation that higher ranks ($r=16$) produced inferior results, despite increased capacity, suggests overfitting to the training data, highlighting the importance of appropriate regularization through rank selection.

The dramatic performance differences between sampling strategies underscore the importance of selecting the appropriate generation methodology. Greedy decoding's severe repetition issues (maximum repetition: 5) and low diversity (distinct-1: 0.347) confirm findings from Holtzman et al. (2020) regarding neural text degeneration. Nucleus sampling's superior performance (distinct-1: 0.949, distinct-2: 1.000) shows that appropriate probability distribution truncation effectively prevents degeneration while maintaining coherence.

The 71% toxicity reduction achieved through combined fine-tuning and nucleus sampling represents a significant advancement in safe text generation. This finding suggests that generation quality and safety are complementary rather than competing objectives, with diverse generation naturally reducing problematic output patterns. The integration of real-time toxicity detection provides practical safeguards for deployment, addressing the ethical concerns raised by Bender et al. (2021) regarding the responsible deployment of AI.

Challenges and Limitations

Several challenges emerged during implementation:

- Computational constraints necessitated the use of GPT-2 Small rather than larger models, which may limit the generalizability of the findings to contemporary large-scale language models.
- Training time optimization required a careful balance between batch size, gradient accumulation, and hardware limitations, with Google Colab Pro providing sufficient but not ideal computational resources.
- Evaluation metrics relied primarily on automated quantitative measures; while statistical significance was established, a comprehensive human evaluation would strengthen conclusions regarding generation quality.
- Domain-specific fine-tuning was not implemented due to time constraints, limiting assessment of LoRA's effectiveness for specialized applications (medical, legal, and scientific domains).

These limitations suggest directions for future investigation while not invalidating core findings regarding parameter-efficient fine-tuning effectiveness and sampling strategy optimization.

Practical Implications

Results provide guidelines for practitioners implementing LLM adaptation: (1) LoRA with rank $r=8$ and learning rate $3e-5$ offers optimal parameter efficiency-performance balance for GPT-2 scale models, (2) nucleus sampling with $\text{top-p}=0.9$ should be default generation strategy for diverse, coherent outputs, (3) real-time toxicity detection is computationally feasible and effective for deployment safety, and (4) attention visualization

provides practical interpretability benefits for model validation and stakeholder communication. The implemented Gradio dashboard effectively demonstrates these principles in an accessible and interactive format, suitable for both research demonstrations and practical deployment scenarios.

Future Work

Several promising directions emerge from this research:

- Scaling experiments to GPT-2 Medium and Large models would verify whether optimal LoRA rank patterns generalize across model sizes, informing configuration selection for contemporary large language models.
- Domain-specific fine-tuning for medical, legal, and scientific applications would assess LoRA's effectiveness for specialized knowledge adaptation, with a focus on accuracy-efficiency trade-offs in high-stakes domains.
- A bias analysis, extending beyond toxicity detection, would address fairness concerns across various demographic dimensions, very relevant for deployment in diverse user populations.
- Human evaluation studies comparing sampling strategies would provide qualitative validation, complementing quantitative metrics, and assessing dimensions such as fluency, coherence, and factual accuracy that are not fully captured by automated measures.
- Investigation of alternative parameter-efficient techniques (adapters, prefix tuning) would enable a comprehensive comparison of fine-tuning approaches, potentially identifying superior methods for specific use cases.

- The development of adaptive sampling strategies that dynamically adjust based on the generation context could further optimize the diversity-coherence balance identified in this research.

These extensions would strengthen theoretical understanding while expanding the practical applicability of LLMs across diverse deployment scenarios, contributing to more accessible, efficient, and responsible adaptation methodologies.

Conclusion

This research demonstrates that parameter-efficient fine-tuning via LoRA enables a substantial reduction in computational requirements (a 99.76% parameter reduction) while maintaining model performance. With an optimal configuration ($r=8$, learning rate $3e-5$), the model achieves a perplexity of 31.89. An evaluation of sampling strategies identified nucleus sampling (top- $p=0.9$) as superior across all diversity metrics (distinct-1: 0.949, distinct-2: 1.000), while achieving a 71% reduction in toxicity compared to the baseline. The integration of attention visualization enhanced model interpretability, and deployment in an interactive Gradio dashboard demonstrated its practical applicability.

These findings address important barriers to LLM adaptation, providing empirically validated guidelines that democratize access to customized language models. The combination of parameter efficiency, safety optimization, and interpretability demonstrates that responsible, accessible AI deployment is achievable through careful methodology selection. As language models continue to advance in capability and scale, these parameter-efficient techniques will become increasingly important for enabling widespread practical applications across diverse domains and organizational contexts.

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